

GOALS AT PRIMARY HEALTH CARE LEVEL

- education of patients on healthy lifestyle issues including nutrition, physical activity and weight control
- effective screening for diabetes with early detection and correct diagnosis
- education of patients, families and health care workers, about diabetes
- appropriate treatment for, and management of all persons with diabetes
- ensuring the permanent availability of oral anti-diabetic agents, insulin, syringes and other disposables, and other medication
- patient self-monitoring to achieve glycaemic control, using appropriate technology and instruction in its use
- prevention, detection and management of complications
- regular oral health promotion and routine oral screening to prevent oral disease, with appropriate referral to oral hygienists for preventive treatment and to dentists for periodontal treatment
- appropriate referral to a higher level of care, and return referral, as required
- appropriate maintenance of patient records and medical history
- regular assessment of effectiveness of services for diabetic patients
- regular assessment of patient's chronic illness coping skills
- patient planning of pregnancies.

SCOPE OF THE POLICY GUIDELINES

TARGET POPULATION

Primary target:

- Adults with Type 2 diabetes
- Family members/care givers of above
- All health professionals at PHC level
- Persons at high risk of developing Type 2 diabetes

Secondary target:

- Health professionals at other levels of care
- Provincial, District and Facility Managers and NCD Managers



A. DEFINITION, SYMPTOMS, DIAGNOSIS, SCREENING, ORGANIZATION OF DIABETES CARE

DEFINITION OF DIABETES MELLITUS ^{2,3,4,5,6}

Diabetes mellitus (diabetes) is a metabolic disorder with heterogenous aetiologies, which is characterized by chronic hyperglycaemia and disturbances of carbohydrate, fat and protein metabolism resulting from defects in insulin secretion, insulin action or both.

Several pathogenic processes are involved in the development of diabetes. These include processes that impair or destroy the function of the pancreatic beta cells, with consequent *insulin deficiency*, and others that result in resistance to insulin action (*insulin resistance/insulin insensitivity*). The abnormalities of carbohydrate, fat and protein metabolism are due to deficient action of insulin on target tissues resulting from insensitivity or lack of insulin or both.

SYMPTOMS OF DIABETES

Characteristic symptoms:	polydipsia, polyuria, weight loss
Other symptoms:	blurring of vision
	general body weakness and lethargy
	susceptibility to infections, especially skin,
	urogenital tract and respiratory tract
	pruritus vulvae
	poor wound healing

Symptoms may not be severe, or may be absent.

DIAGNOSIS OF DIABETES AND OTHER CATEGORIES OF INTERMEDIATE GLYCAEMIA ^{2,3,4,5,6}

For clinical purposes, the diagnosis of diabetes should always be confirmed by repeating the same test on another day.

The diagnosis of diabetes should be based on formal laboratory test results (not point of care tests) prior to the initiation of any therapy.

Table 1 Criteria for the diagnosis of diabetes and categories of intermediate hyperglycaemia

Diagnostic test	Impaired Fasting Glucose (IFG)	Impaired Glucose Tolerance (IGT)	Diabetes
Random plasma glucose (RPG) ⁱ	–	–	≥ 11.1 mmol/L if classic symptoms of diabetes or hyperglycaemic crisis ⁱⁱ is present; or
Fasting plasma glucose (FPG) ⁱⁱⁱ	6.1 – 6.9 mmol/L	< 7.0 mmol/L (if measured)	≥ 7.0 mmol/L; or
Two-hour plasma glucose (2-h PG) during oral glucose tolerance test (OGTT) ^{iv}	< 7.8 mmol/L (if measured)	7.8 – 11.0 mmol/L	≥ 11.1 mmol/L; or
Glycated haemoglobin A _{1c} (HbA _{1c}) ^v	–	–	≥ 6.5%

ⁱ Random (casual) – any time of day without regard to time of last meal.

ⁱⁱ Hyperglycaemic crisis - refers to diabetic ketoacidosis or hyperosmolar non-ketotic hyperglycaemia

ⁱⁱⁱ Fasting - no caloric intake for at least 8 hours

^{iv} OGTT - as per WHO recommendation using a glucose load of 75g anhydrous glucose (or 90g monohydrous glucose) dissolved in 250ml water, ingested over 5 minutes

^v HbA_{1c} – provided the test method meets stringent quality assurance, that the assay is standardized according to criteria aligned with the international reference values [National Glycohemoglobin Standardization Program (NGSP) – certified and standardized to the Diabetes Control and Complications Trial (DCCT) assay], and that there are no conditions present which preclude its accurate measurement.

Table 2 Use of HbA_{1c} in the diagnosis of diabetes mellitus

Diagnosis of diabetes	- HbA_{1c} ≥ 6.5% - HbA_{1c} < 6.5% does not exclude diagnosis by blood glucose - Glucose-based tests (FPG, OGTT) are still valid
Interpretation of HbA _{1c} < 6.5%	No recommendation because of insufficient evidence
Provisos for use of HbA _{1c} for diagnosis	- Stringent quality assurance tests in place - Assays standardized to criteria aligned with international reference values - Low cost and wide availability - No conditions present which preclude accurate measurement
Choice between HbA _{1c} and plasma glucose should be based on local considerations	- Cost - Availability of equipment - National quality assurance system - Population characteristics (prevalence of malaria or haemoglobinopathies) - Accurate blood glucose measurement must generally be available at primary health care level before introducing HbA_{1c} measurement as a diagnostic tool

1. Diagnosis in symptomatic individuals and unequivocal hyperglycaemia (Table 1)

In patients with symptoms of diabetes (polyuria, polydipsia, weight loss) or unequivocal hyperglycaemia (i.e. hyperglycaemic crisis: diabetic ketoacidosis or hyperosmolar non-ketotic hyperglycaemia) – a single abnormal test is sufficient to confirm the diagnosis of diabetes.

However, even severe hyperglycaemia detected under conditions of acute infective, traumatic, cardiovascular or other stress, may be transitory and should not be regarded as diagnostic of diabetes until subsequently confirmed.

2. Diagnosis in asymptomatic individuals or doubtful hyperglycaemia and high risk individuals

- Fasting plasma glucose ≥ 7.0 mmol/L

OR

- 2-h plasma glucose ≥ 11.1 mmol/L during 75 g OGTT

A 75g OGTT is indicated in the following circumstances:

- High-risk individuals (refer Table 3 below)
- If at initial testing

FPG: 5.6 - 6.9 mmol/l

RPG: 5.6 - 11.0 mmol/l (testing for FPG instead of OGTT also acceptable)

Diagnosis will be more difficult in asymptomatic persons or persons about whom there is doubt about the presence of persistent hyperglycaemia – a single abnormal test result may not be used to diagnose diabetes. The same test must be repeated on another day to confirm diagnosis. Two abnormal results, using the same test method, will confirm the diagnosis of diabetes.

REFER Annexure E – Glycated Haemoglobin

SCREENING FOR EARLY DETECTION OF TYPE 2 DIABETES IN ADULTS ^{2,3,4,5,6}

Screening should only be carried out **within a health care setting**, because of the need for follow up. As previously stated, with the increasing global prevalence of diabetes, there is a major need for improved screening.

Community screening **outside** a health care setting (fun days, shopping centres) is not recommended because individuals with abnormal/positive tests may not seek, or have access to appropriate follow-up testing and care. For those who test negative, there may be failure to ensure appropriate repeat testing. Community screening may also be poorly targeted – it may fail to reach groups most at risk and inappropriately test those at low risk or even those already diagnosed.

Random screening for all adults is not recommended until after the age of 45 years.

Notwithstanding the above, diabetes may be identified in the following instances:

- low-risk individuals undergoing incidental glucose testing (random screening) or
- individuals identified as being at high-risk during routine consultations for unrelated health matters (opportunistic screening)

OR

- individuals deliberately identified and tested because of their high-risk status (targeted screening).

Table 3 Criteria for screening for type 2 diabetes in asymptomatic adults

Indications	High-risk individuals: All adults, any age, with BMI ≥ 25 kg/m² (overweight or obese) + 1 or more additional risk factors for future diabetes • physical inactivity • hypertension (BP $\geq 140/90$ mmHg) • family history of diabetes (1st degree) • dyslipidaemia (serum high-density lipoprotein [HDL] cholesterol < 0.90 mmol/L or triglycerides > 2.82 mmol/L) • polycystic ovarian syndrome • high-risk ethnic group (those of South Asian descent) • cardiovascular disease history • history of GDM (gestational diabetes mellitus) or baby > 4 kg at birth • IFG or IGT (previously diagnosed) • other conditions associated with insulin resistance If no risk factors: Age ≥ 45 years
Frequency	At 3 year intervals if tests are normal More frequently, based on initial result and risk status (annually in those with IFG, IGT, or those with multiple risk factors)
Test method	FPG, 2 hour PG (OGTT) or HbA_{1c} OGTT is the preferred test in high-risk individuals

ORGANIZATION OF DIABETES CARE ^{4,5}

As diabetes is a complex disorder, a systematic approach to the organization of care is essential. The elements in this approach include well-trained and dedicated personal; calibrated and functioning equipment; management and referral protocols; regular and sustainable supply of medication, syringes and consumables; a register of all patients to facilitate recall for non-attendance/contact for specific aspects of regular care; legible patient records; flow charts and annual review charts.

A process of regular audit needs to be instituted, to review implementation of interventions, so as to improve standard of care and service delivery.

Requirements at Clinic Level for a Diabetes Service

- appropriately trained personnel, dedicated to the management of chronic diseases
- adequate space:
 - for individual consultation and patient privacy
 - for group education
- Protocols and guidelines covering:
 - screening
 - regular care
 - referrals
- Basic PHC equipment as per standards and norms, including
 - tape measure (for waist circumference)
 - scale for weight
 - height measure
 - accurate and regularly calibrated BP machines including 3 cuff sizes

- Monofilament or tuning fork
- Blood glucose meters – regularly calibrated and in working order with the appropriate test strips
- HbA_{1c} testing equipment, to enable on site testing
- fundoscope
- Snellen chart
- Uninterrupted supply of medication, syringes, consumables
- Register with recall system for non-attendees
- Annual audit of:
 - numbers of patients reaching targets for controlled glycaemia, blood pressure and lipids (indicator will utilise the guideline cutpoints)
 - numbers of patients receiving designated processes of care, as per guidelines (i.e. each procedure would be listed on a YES/NO tick list (e.g. BP checked at each visit; HbA_{1c} done; record of foot exam; retinopathy assessed) and a yes/no would be recorded. A certain number of randomly selected patient records would be reviewed. The % of patients from the random sample that had had all procedures performed would be reported.
- Education material
- Support groups



Schedule of visits to a diabetes clinic

The frequency of examinations and tests are based on the assumption that the last set of observations was normal. In the presence of abnormalities, the frequency must be increased.

Table 4 Patient history, physical examination, bio-chemistry and other activities recommended during visits to a diabetes clinic

INITIAL VISIT		
HISTORY - Symptoms of hyperglycaemia and duration - Relevant family history - Other risk factors: - high birth-weight baby - gestational diabetes - Relevant medical history - co-morbid conditions - symptoms of complications * Cardiovascular; neurological * bladder and sexual functions (i.e. erectile dysfunction) * Feet/vision * Infections - Medication history - current medicine use - allergies to medication - Symptoms of hypoglycaemia - Vaccinations - Pneumococcal (date) - Influenza (date) - Lifestyle - weight history - physical activity - eating pattern - tobacco smoking - alcohol consumption - Psycho-social - depression - occupation - family/community support - Home monitoring chart (if relevant)	PHYSICAL EXAMINATION - Weight & height & BMI kg/m² - Waist circumference (cm) - Blood pressure (mmHg) - Feet - Inspection: soft tissue injuries, ulcers, infections, - deformities, Footwear - monofilament assessment - vibration sense using tuning fork, or pinprick sensation - Ankle jerk - peripheral neuropathy: - foot pulses - Oral cavity - dental caries - gum disease - Eyes - visual acuity - direct fundoscopy (dilated pupils) indirect fundoscopy, or fundus photographs - Cardiovascular system - Injection sites (if appropriate) - ECG: for patient with known ischaemic heart disease, >45 years, other CVD risk factors	BIOCHEMISTRY - Blood - glucose - HbA_{1c} - lipids (TC, HDLC, LDLC, TG) - creatinine; calculate eGFR - potassium - HIV - Urine - protein: (albumin/creatinine ratio) OTHER ACTIVITIES - Education - Lifestyle and behavioural changes including smoking cessation - self-management - nutrition advice, eating habits and meal planning - Goal setting - Pre-conception counselling/family planning (as appropriate) - Medication revision/adjustment - Immunisations

3-6 MONTHLY VISITS UNTIL TREATMENT GOALS ARE ACHIEVED

HISTORY - Symptoms of hyperglycaemia and duration - Medication – assess adherence and side-effects - Symptoms of hypoglycaemia - Lifestyle (as at initial visit) - Depression - Review home monitoring chart	PHYSICAL EXAMINATION - Weight & BMI - Waist circumference - Blood pressure - Feet - Inspection: soft tissue injuries, ulcers, infections, deformities, Footwear - Injection sites (if applicable)	BIOCHEMISTRY - Blood – glucose – HbA_{1c} – 3-6 monthly if NOT at target and when changing medication OTHER ACTIVITIES As at initial visit especially medication review
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ANNUAL VISIT

HISTORY As at initial visit	PHYSICAL EXAMINATION As at initial visit Screening for retinopathy: 2 yearly if last 2 examinations were normal; - more frequent examinations in the presence of abnormalities	BIOCHEMISTRY As at initial visit OTHER ACTIVITIES As at initial visit
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Table 5 Monitoring of patients with diabetes and frequency of tests

TESTS/PROCEDURES	FREQUENCY OF TESTS				
	First visit	Each Visit	3 monthly	6 monthly	12 monthly
Blood glucose (finger prick)	•	•			
HbA _{1c} – at target; no change in treatment	•				•
HbA _{1c} – above target or medication has been adjusted	•		• or	•	
Blood Pressure	•	•			
Lipid profile • abnormal Every diabetic >40 years will receive 10 mg statin irrespective of baseline	•		•		•
Injection sites	•		•		
Self-monitoring technique				•	
Weight and waist	•	•			
BMI	•		•		
Foot and shoe inspection (Refer section C7)	•	•			
Foot examination (Refer Table 11)	•				•

TESTS/PROCEDURES	FREQUENCY OF TESTS				
	First visit	Each Visit	3 monthly	6 monthly	12 monthly
Protein Albumin/creatinine ratio	•				•
SCREENING FOR CKD: 1) Random 20ml urine sample for albumin-to-creatinine ratio (ACR) Normal range: Men: <2mg/mmol Women: <2.8mg/mmol If abnormal , repeat twice within the following 3 months 2) 10ml non-fasting clotted blood for Serum creatinine concentration, for conversion into the eGFR Normal: > 60 ml/minute If abnormal , repeat once within the following 3 months	•				•
Eye examination for retinopathy (more frequently if there is blurring, loss of visual acuity, significant retinopathy)	•				•
Referral to health educator and/or dietician (more frequently if out-of-control, overweight or obese)	•		•		•
ECG: known ischaemic heart disease, > 45 years; other CVD risk factors.	•				•

Glycosuria is a poor measure of glycaemic control.

If this is the only tool available, then test the second voided specimen of the day.



B. MANAGEMENT OF BLOOD GLUCOSE

TARGETS FOR GLYCAEMIC CONTROL ^{8,9,10}

Blood glucose targets:

1. Fasting plasma glucose (FPG)

The "normal" FPG is accepted as 6.1 mmol/L, however, the recommended fasting or pre-prandial target is 4.0 - 7.0 mmol/L

2. Post-prandial glucose (PPG)

In non-diabetic individuals, peak PPG generally does not exceed 7.8 mmol/L. The actual target for persons with diabetes will depend on the target HbA_{1c} (see Table below)

HbA_{1c} targets:

Epidemiological evidence shows that persons with type 2 diabetes, with HbA_{1c} levels > 7.5%, have a 2.5- to 5-fold greater relative risk of developing microvascular complications and a 5-fold greater risk of developing peripheral artery disease. HbA_{1c} targets must be individualized (see Table below)

Table 6 Targets for fasting plasma glucose, post-prandial glucose, HbA_{1c} in different patient types

Patient type	Target FPG	Target PPG	Target HbA _{1c}
Majority of patients	4.0 – 7.0 mmol/L	5.0 – 10.0 mmol/L	< 7 %
Young Low risk Newly diagnosed No cardiovascular disease	4.0 – 7.0 mmol/L	4.4 -7.8 mmol/L	< 6.5 %
Older persons High risk Hypoglycaemic unaware Poor short-term prognosis	4.0 – 7.0 mmol/L	< 12 mmol/L	< 7.5 %

Very important: Young diabetics < 30 years should be referred for hospital level assessment at least once per year.

GLYCAEMIC MANAGEMENT OF TYPE 2 DIABETES ³

1. Non-pharmacological management:

REFER: **Annexure A1 - Diabetes Education**
Annexure A2 - Lifestyle Modification

2. Pharmacological management:

HbA_{1c} < 7% is the primary target for glycaemic control

There are 3 main types of hypoglycaemic agents available for use - these are biguanides, sulphonylureas (SU), and insulin.

Biguanides: Metformin is the only product available and is the 'anchor' oral anti-diabetic agent in the management of type 2 diabetes.

About 30% of users will report gastro-intestinal side-effects, which can be minimised by titrating the dose gradually over one or two months, or by temporarily discontinuing the drug before re-introducing it. When used as monotherapy, metformin can reduce HbA_{1c} by 1 – 2 %.

In patients with severe renal impairment, adjust metformin dose according to the table below:

eGFR	Action
> 30 – 60 ml/minute	Continue, use 50% of dose (maximum 500 mg 12 hourly) Increase monitoring of renal function (every 3 – 6 months)
< 30 ml/minute	Stop metformin

Sulphonylureas: Major adverse effects include hypoglycaemia (the most serious adverse effect) especially in older persons and in the presence of poly-pharmacy and weight gain (average weight gain of 5.3 kg over the first 6 years with most of the weight gain occurring in the first year of treatment). HbA_{1c} reduction ranges from 1.5 – 2 %

Insulin: Patients with type 2 diabetes often require insulin to achieve glycaemic control, thus insulin may be added to other therapies, as second- or third-line therapy.



Algorithm 1: Non-insulin and insulin therapy**STEPS 1 – 3 TO ACHIEVE EFFECTIVE GLYCAEMIC MANAGEMENT**

All the required tests at the initial visit, must be completed.

ENTRY TO STEP 1

- At diagnosis of diabetes mellitus, after the required tests have been completed, including serum creatinine/eGFR.
- If patient is thin (BMI <20kg/m²), refer to hospital level care.

STEP 1

Lifestyle modification + metformin (biguanide) for 3 months

STEPS	TREATMENT AND DURATION	TARGET	CAUTION	CONTRA-INDICATIONS
Lifestyle modification	For life • provide appropriate eating plan; • provide weight loss plan until ideal weight is reached • assess monthly until stabilised	HbA_{1c} < 7 Body weight reduced by 5 -10%		
Metformin, oral	500mg daily x 2 weeks 500mg 12 hourly x 2 weeks 850mg 12 hourly X 2 weeks 850mg 8 hourly	MAX: 2550mg daily, in 2 or 3 divided doses	Use with care in the elderly if serum creatinine is > 130 µmol/L Gastro-intestinal side effects in ± 30% of users may limit compliance. Pregnant patients should be managed at a specialist level	• Severe renal impairment (see "biguanides" above) • Severe liver disease • Use of intravenous contrast media • Major surgical procedures • Congestive heart failure • Acute myocardial infarction

Failed step 1 (HbA_{1c} > 7% and < 8.5% or FPG > 8 mmol/L, despite lifestyle modification and adherence to treatment with maximal dose of metformin for 2-3 months).

Continue lifestyle modification + metformin and add sulphonylurea (gliclazide or glibenclamide)

Before adding the sulphonylurea check dietary and medicine regimen adherence.

Prescribe second oral agent incrementally.

STEPS	TREATMENT AND DURATION	TARGET	CAUTION	CONTRA-INDICATIONS
Lifestyle modification + metformin at maximum dose + Sulphonylurea a) Gliclazide, oral - every dose to be taken with meals OR b) Glibenclamide, oral - every dose to be taken with meals	Continue 40mg daily with breakfast Increase dose by 40mg daily at 2-weekly intervals. If more than 80mg daily is needed, divide total daily dose in two 2.5mg daily with breakfast Increase dose by 2.5mg daily at 2-weekly intervals. If 7.5mg daily or more is needed, divide total daily dose into two, with larger dose in the morning	HbA_{1c} < 7 Body weight further reduced, if necessary MAX: 320mg daily, in two divided doses MAX: 15 mg daily, in two divided doses	Hypoglycaemia occurs in > 11% of patients. Older persons Causes weight gain Persons with porphyria During lactation Glibenclamide has a long duration of action and is associated with a high rate of hypoglycaemia Older persons – risk of hypoglycaemia increases exponentially with age (gliclazide preferred)	• Brittle or unstable diabetes • Renal impairment (eGFR < 60 ml/minute) • Severe liver dysfunction • Allergy to sulphonamides Sulphonyureas are ABSOLUTELY contra-indicated in renal impairment - eGFR has not been measured in the preceding year or it is < 60 ml/minute/1.73m²

ENTRY TO STEP 3

- Failed step 2: ($HbA_{1c} > 7\%$ and $< 8.5\%$ or $FPG > 8 \text{ mmol/L}$, despite lifestyle modification and adherence to combination oral medication after 3 months)

STEP 3 - INSULIN THERAPY

Continue lifestyle modification + metformin + insulin (sub-cutaneous injection)

Discontinue sulphonylurea if patient is placed on biphasic insulin.

Continue with metformin, especially in the overweight/obese patient because it reduces insulin resistance and thus reduces the insulin dosage.

STEPS	TREATMENT AND DURATION	TARGET	CAUTION	CONTRA-INDICATIONS
Lifestyle modification + Metformin + Sulphonylurea + Insulin (either basal or pre-mix) + Education on insulin therapy Refer: Self Monitoring Blood Glucose (SMBG)	Continue Continue Discontinue if patient is placed on biphasic insulin		If unexplained hypoglycaemia occurs when using insulin, instruct patient to reduce last injected insulin dose by 10% The current insulin regimen should be re-evaluated if targets are not reached within 3-6 months	

INSULIN TYPE	STARTING DOSE	INCREMENT	MAXIMUM DAILY DOSE	REQUIRED TARGET
Intermediate- (NPH) or long-acting (Maintain all oral medications)	10 units at bedtime, not later than 22:00 (10 p.m.)	Titrate by 2 units every 3 – 7 days until fasting glucose target is reached	20 units <i>Ensure that patient is adherent before increasing insulin dose (See Annexure F)</i>	Fasting glucose is 4-7 mmol/L If control is not achieved and $HbA_{1c} > 7\%$, then change to Biphasic insulin
Biphasic (Discontinue all oral medication <u>except</u> Metformin)	Starting dose: total daily insulin – 15 units/day Administer $\frac{2}{3}$ of the total dose (10 units) before breakfast, and $\frac{1}{3}$ (5 units) before supper	Titrate by 4 units per week until fasting glucose target is reached - First increment is added to morning dose - Second increment is added to evening dose	30 units <i>Ensure that patient is adherent before increasing insulin dose (See Annexure F)</i>	Fasting glucose is 4-7 mmol/L If control is not achieved and $HbA_{1c} > 7\%$ - REFER

ENTRY TO STEP 4 – HOSPITAL LEVEL CARE

- presence of severe decompensation (includes **any** of the following: $FPG > 15 \text{ mmol/L}$, $HbA_{1c} > 11\%$, marked polyuria & polydipsia, weight loss $> 5\%$ or ketoacidosis).
- thin patients ($BMI < 20 \text{ kg/m}^2$) or ill patients
- patients with severe infections; undergoing major surgery; in the presence of severe hyperglycaemia ($FPG > 20 \text{ mmol/L}$) or DKA, and pregnancy.
- severe liver or kidney failure

General notes when patients are started on insulin:

- Patients must be given structured education on types of insulin; injection technique; injection sites (abdomen, thighs, arms are recommended) ; insulin storage.
- Patients should be informed of increased risk of weight gain.
- Patients should be aware of the significant risk of hypoglycaemia.
- The insulin regimen must be matched with the patient's lifestyle, acceptance and numeracy skills.
- Patients must be given **written** instructions for insulin dose titration – this titration must be ongoing.
- Patients should monitor their finger-prick glucose as per the written instructions and **MUST** be provided with sufficient glucometer strips for their needs.
- Patients must be counselled on 'sharps disposal' and should use a clinic or pharmacy for this purpose.
- Patients should be advised on frequency of meals and snacks, according to their individual insulin regimen.
- Patients should be informed about possible acute complications (hypoglycaemia and hyperglycaemia), and how to recognise and manage them.
- Re-evaluate the current insulin regimen if targets are not reached within 3 – 6 months.
- If glycaemic targets are not met with basal or biphasic insulin then intensive insulin therapy (with multiple daily injections) must be considered by the Doctor.
- Specialist referral is appropriate at any stage if glycaemic targets remain unmet.

SELF-MONITORING OF BLOOD GLUCOSE (SMBG) :

- SMBG will encourage 'qualifying' patients to assume a greater responsibility for glycaemic control, thereby improving confidence and self-management.
- Self-monitoring technique should be checked 6 monthly by the health care team.
- Increased testing will be required with the development of
 - acute illness
 - periods of poor glycaemic control
 - frequent hypoglycaemic episodes
 - pregnancy
 - adjustments to therapy

Self-monitoring for individuals on oral agents

- there is a lack of evidence to support the use of self monitoring with glucose tests strips in patients on oral agents.

Self-monitoring for individuals on insulin:

SMBG is essential, but the frequency depends on the insulin regimen being used.

Persons injecting insulin 2 – 4 times a day:	Testing should be undertaken at least twice daily. When stable this can be reduced to once daily.
Persons injecting insulin once daily, with or without oral agents:	Once daily testing at variable times is recommended

Intermediate- or long-acting insulin:

- Patient must monitor fasting (on waking) finger-prick glucose daily.
- Patient must calculate average fasting glucose every 3 – 7 days

Biphasic insulin:

- Patient must monitor his/her finger-prick glucose before breakfast and before supper

A glucometer with sufficient test strips **must** be available to all patients on insulin therapy.

'SICK DAYS' - INFORMATION FOR PATIENTS DURING PERIODS OF ILLNESS OR STRESS

During illnesses or times of stress, or when "out of control" for any reason, diabetic patients may require additional insulin, or escalation of therapy to include insulin therapy.

Additional insulin, with appropriate Instructions, should only be given to patients who are able to test their glucose at home and who will be able to carry out the instructions safely and correctly when their blood glucose is raised.

- Additional doses of insulin: Only **short acting** insulin is used, which should be added to the usual insulin doses and injected **before each meal**. This can be done in **one** of the following ways:
 - increase the dose of the short acting insulin by **10% of the total daily insulin dose**,
- **OR**
 - increase the usual short acting dose by 20%
- Do not discontinue the usual insulin doses.
- Continue oral agents.
- If vomiting or dehydrated, STOP metformin and seek medical advice or go to the clinic/hospital **urgently**.
- If patient is vomiting, or if consciousness is impaired, **urgent** assistance should be sought from a doctor, clinic or hospital.
- If unable to reduce blood glucose levels to resolve the hyperglycaemia, seek medical advice or go to clinic/hospital **urgently**.
- **All insulin patients** with blood glucose levels sustained **above 15mmol/L** despite **two** correcting doses of insulin, require medical review.
- **All non-insulin patients** staying **above 15mmol/L** over a 4 hour period, require medical review.

Food: Patients should keep to their usual diet. If this is not possible, small frequent helpings of soft foods or liquids containing carbohydrates (starch) should be taken e.g. porridge, soup.

Fluids: Patients should drink plenty of water or diet mineral drinks.

Nausea: When patient is feeling nauseous and not tolerating solids, he/she should drink unsweetened drinks. Small quantities of sweetened drinks (fruit juice) should also be included to prevent hypoglycaemia.

Blood glucose: Should be checked before every meal and at bedtime

Regular follow-up appointments and the institution of a patient recall system are fundamentally important in achieving therapeutic objectives.

The aims of follow-up are to:

- Monitor and optimise glycaemic control, blood pressure and blood lipids.
- Monitor weight.
- Assist and motivate patient to initiate and continue self-care and lifestyle modification.
- Re-evaluate self-monitoring techniques and injection techniques
- Provide therapeutic education and re-education.
- Detect and manage long-term complications at an early stage.
- Assess compliance and drug side effects
- Assess interaction of disease and treatment with social factors
- Advise on treatment and referral if necessary.

REFERRALS

1. Urgent Referrals – same day

- Metabolic
 - dehydration and hypotension
 - nausea, vomiting and abdominal pain
 - stupor, confusion, pre-coma
 - hyperglycaemia >25 mmol/L
 - hypoglycaemic coma unresponsive to intravenous glucose therapy
 - DKA
- Severe complications
 - sepsis, infections with a slow onset of progressive apathy leading to confusion, stupor, pre-coma and coma
 - gangrene
 - sudden deterioration in vision
 - serious infections (TB, urinary tract infections, abscesses/boils, pneumonia)
- Oral health complications

Start sodium chloride 0.9% infusion before transfer of ill patients – **do not** administer insulin during transportation.

2. Standard Referrals (as per guidelines)

- Thin patients (BMI <20kg/m²) or ill patients
- Patients with severe infections
- Patients undergoing major surgery
- Pregnant women.

- Failure to optimise diabetes control with insulin [$\text{HbA}_{1c} > 7.0\%$]
- All type 1 diabetics
- Pregnancy with glycosuria/hyperglycaemia/poor obstetric history.
- High-risk feet, including persistent infections and ulceration of the foot.
- Persistent proteinuria of ANY degree. Exclude common causes.
- Undiagnosed visual difficulties/disturbances. Retinopathy on fundoscopy.
- Patients with hepatic and renal dysfunction.

3. Back-referrals

Comprehensive back-referral to primary care level is mandatory.

HYPOGLYCAEMIA ^{11,12}

Hypoglycaemia is a serious and common complication of sulphonylurea and insulin treatment, and may manifest during sleep.

Hypoglycaemia is recognised when plasma glucose falls to < 4.0 mmol/L.

Symptoms may occur at higher values when glucose levels decrease rapidly. Low finger prick blood glucose levels should be confirmed with laboratory measurements, but treatment should not be withheld while awaiting results - hypoglycaemia should be managed immediately.

Complications: Severe hypoglycaemia may cause permanent neurological damage or brain death, and may be responsible for sudden death (the "dead in bed" syndrome) related to cardiac arrhythmia.

Possible causes of hypoglycaemia in patients with diabetes.

- increased exercise (common)
- decreased food intake (common) (missed, late, small meals)
- inappropriate insulin dose
- inappropriate sulphonylurea dose (especially glibenclamide)
- progressive renal failure (causing decreased insulin clearance)
- alcohol intake

Presenting features (symptoms and signs) of hypoglycaemia:

Symptoms	Signs
hunger sweating palpitations tremor tingling sensation anxiety weakness faintness headache	change in behaviour confusion seizures (convulsions) coma

Treatment of Mild Hypoglycaemia

Patients who are aware of the signs and symptoms of hypoglycaemia will be able to relieve the condition early by taking glucose or 2 - 4 teaspoonfuls of sugar with a little water. If necessary, this step should be repeated within 10 - 15 minutes. Thereafter, slowly digestible carbohydrates (e.g. bread) and protein (e.g. milk) should be taken for prolonged restoration of blood glucose.

Glucose/sucrose should **always** be carried by patients in pocket/handbag for possible emergencies.

Treatment of Severe Hypoglycaemia (requires the intervention of another person)

Treat patient immediately. Arrange admission to hospital once patient has recovered.

The following steps should be taken on presentation of a suspected hypoglycaemic patient:

1. Establish a large bore intravenous line.
2. Immediate rapid intravenous injection of up to 50ml of 50% Dextrose solution. Assess clinical **and** biochemical response over the next 5 -10 minutes.
3. If blood glucose remains < 4.4 mmol/L give a second IV injection of 20 - 50ml 50% dextrose.
4. Continue IV infusion of 5% Dextrose in Water, at a rate of about 1 litre in 6 hours (to prevent recurrent hypoglycaemia, particularly if induced by long acting insulin and/or sulphonylurea agents).

NOTE: For patients that are alcoholic, malnourished, mal-absorbers etc. – continue IV infusion with 5% Dextrose in Water + Thiamine 100mg IMI

5. Once blood glucose is normal or elevated, and the patient is awake, provide patient with a snack.
6. If the **patient has not regained consciousness** after **30 minutes**, despite a normal or elevated blood glucose, other causes of coma will need to be considered (e.g. meningitis). **Urgent referral to hospital is indicated.**
7. Refer all hypoglycaemic patients to hospital for observation and education to prevent further hypoglycaemic episodes.
8. In hospital, monitor the clinical state and blood glucose 4-hourly for 24 - 48 hours.
9. Always try to identify the underlying cause of the hypoglycaemic episode.

Important points to remember:

- If hypoglycaemia was caused by a sulphonylurea, the patient may need hospitalisation and IV dextrose/glucose infusion for several days - especially if glibenclamide was the cause.
- Honey or syrup may be rubbed in the gums of patients losing consciousness if other medication is not readily at hand.

Patient education:

All diabetic patients must receive education on the recognition and treatment of hypoglycaemia.

It is essential that the cause of each hypoglycaemic episode is established and appropriate preventive action taken to prevent further episodes.

Recurrent hypoglycaemia:

Patients who experience recurrent severe hypoglycaemic reactions should be given more intensive adherence and monitoring counselling.

All patients with recurrent hypoglycaemia should be referred to hospital for assessment.

Consider the following in cases of recurrent hypoglycaemia:

- inappropriate management;
- poor adherence to treatment regimen and management plan;
- alcohol abuse;
- self-induced hypoglycaemia
- the advent of renal failure
- hypoglycaemia unawareness.

Hypoglycaemia awareness:

Recurrent hypoglycaemia may be the cause **or** consequence of hypoglycaemic unawareness.

This complication occurs after some years of diabetes and is associated with hypoglycaemia-associated autonomic failure. These patients do not exhibit the early symptoms of hypoglycaemia, and consequently present with neuroglycopenia (confusion, seizures, coma).

Evidence exists that in some of these individuals scrupulous avoidance of hypoglycaemia for several weeks can reverse hypoglycaemic unawareness.

HYPERGLYCAEMIC EMERGENCIES ^{13,14,15}

(1) Diabetic ketoacidosis (DKA)

(2) Hyperglycaemic hyperosmolar state (HHS)

DKA and HHS should be suspected whenever patients have significant hyperglycaemia, especially if they are systemically unwell or are known to have diabetes.

These conditions have significant morbidity and mortality, and emergency treatment with intravenous fluids and insulin is essential.

1. DIABETIC KETOACIDOSIS [DKA]

DKA is characterised by uncontrolled hyperglycaemia, metabolic acidosis and increased total body ketones. In South Africa, DKA carries a higher mortality than in the developed world. It can present at any age although it is more common in young patients.

An acute DKA episode usually develops within 24 hours

2. HYPERGLYCAEMIC HYPEROSMOLAR STATE [HHS]

Hyperglycaemic hyperosmolar state is characterised by the slow development of marked hyperglycaemia (usually > 50 mmol/L), hyperosmolality and severe dehydration. Ketonuria may be slight or absent. The condition usually affects middle-aged or older patients and carries a high mortality.

HHS usually evolves over several days to weeks.

For both DKA and HHS the classical clinical picture includes a history of

polyuria,

polydipsia,

vomiting,

dehydration,

weight loss,

weakness and

change in mental status (can vary from mild confusion to profound lethargy or coma, the latter is more common in HHS).

Focal neurological signs and seizures may occur in HHS.

Patients with DKA commonly complain of diffuse abdominal pain. Physical findings may include Kussmaul breathing (in DKA only) or shock, with hypotension and tachycardia.

History taking and examination should not delay the initial investigations, management and referral to hospital.

Precipitating factors of DKA and HHS

- Infection is the most common precipitating factor in DKA and HHS.
- Other precipitants include:
 - discontinuation of insulin,
 - myocardial infarction (MI) or
 - cerebrovascular accident (CVA).
- In elderly patients, restricted water intake due to the patient being bedridden can lead to dehydration and precipitate HHS. Drugs such as diuretics may exacerbate the condition.
- In addition, new-onset diabetes commonly leads to DKA in patients with type 1 diabetes and HHS in patients with type 2 diabetes mellitus.
- In a significant proportion of patients, no precipitant may be identified.

Treatment of Hyperglycaemic Coma at Primary Care Level

Rapid treatment **and referral** is essential.

The following steps should be followed:

- Confirm diagnosis
- Check capillary glucose and urine ketones to confirm hyperglycaemia and ketonuria.
- Be cautious in renal and cardiac disease.

- In the absence of renal or cardiac compromise, administer intravenous (IV) fluids, starting with normal saline, 15–20 ml/kg in the first hour
 - Subsequent infusion rate: 10 ml/kg/hour with 20 ml/kg boluses if shocked.
 - Do not exceed 50 ml/kg in the first 4 hours.
 - Correct estimated deficits over 24 hours.

In the older patient with suspected HHS or cardiac compromise, change to 0.45% saline after the first litre of normal saline.

- Insulin

When referral will take more than 2 hours and a diagnosis of diabetes with hyperglycaemia is confirmed, administer short acting insulin, IM, 0.1 unit/kg.

- When giving insulin IM, do not use insulin needle.

CAUTION

Do not administer IV short-acting insulin if the serum electrolyte status, especially potassium, is not known.

Continue with IV fluids but delay giving insulin in these cases in consultation with referral facility as this delay should not negatively influence the patient, but hypokalaemia with resultant cardiac dysrhythmias definitely will.

See section 8.6.2: Diabetic Ketoacidosis (DKA) and Hyperosmolar Nonketotic Diabetic Coma (HONK) of the STG and EML for Hospital Level (Adult), 2012 edition

Refer urgently with drip in place and running at planned rate.

Table 7: Differentiation between the types of hyperglycaemic coma

	DIABETIC KETOACIDOSIS (DKA)	HYPERGLYCAEMIC HYPEROSMOLAR COMA (HHS)
History	Known T1 diabetes Newly-diagnosed	Known T2 diabetes Newly-diagnosed
Precipitants	Infection Non-compliance on insulin	Infection Myocardial infarction Cerebrovascular accident diuretic use
Age frequency	Younger patients	Usually older persons
Onset	Hours to days	Days to weeks
Symptoms	Polyuria Polydipsia Anorexia Nausea and vomiting Abdominal pain	Polyuria, Polydipsia, Increasing somnolence
Signs	Kussmaul respiration: deep, sighing breathing Dehydration Confusion Nausea and vomiting	Severe dehydration Mental status changes Focal neurological signs Seizures
Urine ketones	Strongly positive	Positive or negative
Serum ketones	Strongly positive	Negative or only weakly positive
Blood glucose	Raised	Markedly raised
Blood pH	Decreased	Normal or slightly decreased
Serum bicarbonate	Low	Normal or slightly decreased



C. MANAGEMENT OF CO-MORBIDITIES AND COMPLICATIONS

Obesity

Cardiovascular risk and dyslipidaemia

Aspirin therapy

Hypertension

Chronic kidney disease

Retinopathy

Neuropathy and foot problems

C1. OBESITY ^{16,17,18}

The aim is to lose 5 – 10% of body weight; maintain weight loss; prevent weight regain.

The optimal rate of weight loss is 1 – 2 kg per month

Obesity induces insulin resistance and pancreatic beta-cell dysfunction, which over time lead to worsening hyperglycaemia. Abdominal fat is directly associated with an increased risk of obesity-related cardio-metabolic complications.

Type 2 diabetic patients are very prone to weight gain especially after the introduction of sulphonylureas and the commencement of insulin therapy. In addition, other medications (certain beta blockers, anti-depressants, other) prescribed for diabetes co-morbidities are associated with weight gain.

Patients **must** be made aware of this possibility and precautions (such as starting with a low dose and reinforcing lifestyle modification) should be taken to prevent weight gain.

Weight loss should be supported by **'behavioural therapy'** and should involve:

- Developing specific and realistic goals that can be easily measured
- Developing a reasonable plan for reaching those goals and to prevent relapse
- Making incremental changes rather than large changes

The South Africa Demographic and Health Survey, 2003 reported the following percentages for overweight and obese adult persons:

Age	Overweight		Obese	
	Men	Women	Men	Women
Adults 15+ years	21%	28%	8.8%	27%

These percentages infer the potential magnitude of worsening hyperglycaemia and associated complications.

REFER: Annexure A1 – Diabetes Education

Annexure A2 – Lifestyle Modification

National Guideline: Prevention and Management of Overweight and Obesity in South Africa

C2. CARDIOVASCULAR RISK AND THE MANAGEMENT OF DYSLIPIDAEMIA ^{19,20}

It is important to recognise that type 2 diabetes affects both small (microvascular) and large (macrovascular) vessels. Dyslipidaemia is a major contributor to macrovascular disease, or atherosclerosis, which accounts for up to 70% of all diabetic mortality.

Type 2 diabetes should, therefore, be considered a coronary artery disease risk equivalent and dyslipidaemia and other cardiovascular risk factors should be looked for and aggressively treated in every patient with diabetes.

2.1 Macrovascular disease

The risk of CVD in persons with type 2 diabetes is increased 2- to 3-fold in men and 3- to 5-fold in women compared with people without diabetes.

Myocardial infarction and stroke in diabetic patients appear to be more extensive and are likely to be more rapidly fatal than in non-diabetics.

Atherosclerosis is often accelerated and severe in diabetes. Risk factors for accelerated atherosclerosis include chronic hyperglycaemia, in addition to others - race, gender, hypertension, dyslipidaemia, smoking, social deprivation and obesity.

All these risk factors, **except chronic hyperglycaemia**, also affect non-diabetics.

Table 8 summarises the traditional modifiable risk factors for CVD with the recommended target values in diabetes

Table 8: CVD risk factors and targets for patients with type 2 diabetes

Traditional CVD risk factors	Targets
Cigarette smoking	Cessation
DYSLIPIDAEMIA	
Total Cholesterol	<4.5 mmol/L
LDL-Cholesterol *	<1.8 mmol/L
HDL-Cholesterol	Men > 1.0 mmol/L
	Women > 1.2 mmol/L
Triglycerides	< 1.7 mmol/L
OBESITY	
Body Mass Index (BMI) kg/m ²	< 25
Waist circumference	Men < 94 cm
	Men of South Asian descent < 90 cm
	Women < 80 cm
HYPERTENSION	
Systolic blood pressure	< 140 mmHg
Diastolic blood pressure	< 80 mmHg

* The LDL cholesterol goal is < 2.5 mmol/L in patients with type 2 diabetes who meet the following criteria:

- no cardiovascular disease and no chronic kidney disease
- less than 40 years old OR duration of diabetes is less than 10 years
- no other cardiovascular risk factors

2.2 The Metabolic Syndrome

The Metabolic Syndrome, or insulin resistance syndrome, has become one of the major public-health challenges worldwide. Over the past few decades the prevalence of the Metabolic Syndrome has increased exponentially, affecting up to 25% of Western populations. The increase has paralleled the global epidemic of obesity.

The Metabolic Syndrome increases the risk of developing diabetes approximately 5-fold and doubles the risk of atherosclerotic cardiovascular disease.

Characteristic features of the Syndrome include:

- abdominal obesity,
- atherogenic dyslipidaemia (elevated serum triglyceride and a lowered HDL-cholesterol),
- hypertension and
- elevated fasting glucose.

Other ancillary features may include the polycystic ovarian syndrome (PCOS), non-alcoholic fatty liver disease (NAFLD) as well as sleep apnoea.

Table 9: Harmonised criteria for the clinical diagnosis of the Metabolic Syndrome

Measure	Categorical cut points	
Elevated waist circumference	MEN ≥ 94 cm	WOMEN ≥ 80 cm
Elevated triglycerides	≥ 1.7 mmol/L	
Reduced HDLC	< 1.0 mmol/L – males < 1.3 mmol/L - females	
Elevated blood pressure	Systolic ≥ 130 and/or Diastolic ≥ 85 mmHg	
Elevated fasting glucose	≥ 5.6 mmol/L	

The presence of 3 of the 5 risk factors constitutes the diagnosis of the Metabolic syndrome.

Drug treatment specifically targeted at any one of criteria 2-5 makes that criterion positive, even if the measured variable falls below the cut-off.

Lifestyle changes (diet, weight loss, regular physical activity) - the cornerstone of therapy - can delay, or even prevent, the onset of type 2 diabetes in patients with the metabolic syndrome.

2.3 Diabetic dyslipidaemia

Atherosclerosis accounts for up to 70% of all diabetic mortality in White, Coloured and Asian patients, and whilst it is still uncommon in Blacks, it is on the increase.

Lipid disturbances are common in patients with diabetes, and probably contribute to the high incidence of vascular disease. Lipid abnormalities should therefore be looked for and treated in every patient with diabetes.

The most frequently encountered lipid disturbances in type 2 diabetes are increased serum triglycerides and decreased HDL cholesterol.

2.3.1 Goals of therapy

Table 10: Ideal lipid profile for a patient with diabetes

Total Cholesterol	Men Women	< 4.5 mmol/L
LDL-Cholesterol *		< 1.8 mmol/L
HDL-Cholesterol		> 1.0 mmol/L
		> 1.2 mmol/L
Triglycerides		< 1.7 mmol/L

* The LDL cholesterol goal is < 2.5 mmol/L in patients with type 2 diabetes who meet the following criteria:

- no cardiovascular disease and no chronic kidney disease
- less than 40 years old OR duration of diabetes is less than 10 years
- no other cardiovascular risk factors

2.3.2 Monitoring serum lipids

A full fasting (10 hour fast) lipid profile should be performed when first evaluating a diabetic patient.

If the results of the fasting lipid profile are satisfactory, repeat annually, otherwise repeat in 3 months, after the patient has been following an appropriate lipid-lowering diet, weight reduction has been encouraged, glucose control has been established,

and lipid-lowering therapy has been instituted. Three monthly follow-up measurements are required during the titration phase of lipid-lowering therapy.

2.3.3 Non-pharmacological therapy

Diet is the cornerstone of therapy, with particular stress on:

1. Calories: The calorie content of the diet must be adjusted to achieve ideal body weight. Even moderate weight loss (e.g. 5-10%) can be of great value in improving the dyslipidaemia.
2. Fat intake: This should be limited to < 30% of daily energy content. Cholesterol intake should be restricted to < 300 mg per day.
3. Fibre: The fibre content of the diet should be increased.
4. Alcohol: In the presence of obesity and/or hypertriglyceridaemia alcohol should be avoided. Otherwise alcohol should be restricted to 1-2 units per day.

Poor metabolic control is a contributor to diabetic dyslipidaemia - diabetes **must** be under adequate control. It is important to always exclude secondary causes of hyperlipidaemia (e.g. hypothyroidism, renal disease and alcohol abuse).

2.3.4 Pharmacological therapy

Achieving the recommended LDL-cholesterol level is the primary goal of therapy.

Statins are first line agents for lowering LDL-cholesterol in diabetic patients.

Statin therapy should be added to lifestyle therapy, regardless of baseline lipid levels.

Algorithm 2: Statin therapy

GROUP 1

Type 2 diabetic patients

- **with existing** cardiovascular disease (ischaemic heart disease, cerebrovascular disease or peripheral vascular disease)
- with chronic kidney disease (estimated GFR < 60 ml/min/1.73m²)
- older than 40 years, or duration of diabetes > 10 years, with one or more additional cardiovascular risk factors (hypertension; smokes cigarettes; low HDL-cholesterol level; family history of early coronary heart disease; micro- or macroalbuminuria)

Lifestyle modification + statin therapy (regardless of baseline lipid levels)

STATIN	TREATMENT	COMMENT
Lifestyle modification Simvastatin, oral	10mg daily	These patients are at high risk Refer patients on protease inhibitors (lopinavir/ritonavir) for alternate therapy.
GROUP 2 Type 2 diabetic patients • without established cardiovascular disease or chronic kidney disease • under 40 years of age, with duration of diabetes < 10 years, without additional cardiovascular risk factors Assess risk of developing MI of > 20% in 10 years.		
REFER Random cholesterol > 7.5 mmol/L Fasting (14 hours) triglycerides > 10 mmol/L		

C3. ASPIRIN THERAPY IN DIABETES ^{21,22,23}

Aspirin therapy should be initiated by a medical officer.

The recommended daily dose ranges from 75 – 150 mg of aspirin.

For primary prevention:

Aspirin therapy is not beneficial in all patients with diabetes, and is associated with a significant increase in GIT bleeds and, to a lesser extent, intracranial bleeds.

Aspirin is NOT beneficial in younger patients **without** significant risk factors (<5% risk of CVD over next 10years)

- Give aspirin to all patients with diabetes with a history of cardiovascular disease, i.e.:
 - ischaemic heart disease,
 - peripheral vascular disease,
 - previous thrombotic stroke.
- Determine individual aspirin therapy in diabetic patients with an intermediate risk, (i.e. the 10 year risk of CVD is from 5 – 10%). These include men under 50 years and women under 60 years with no cardiovascular risk factors.

C4. MANAGEMENT OF HYPERTENSION IN PATIENTS WITH TYPE 2 DIABETES ^{24,25,26,27}

Patients with type 2 diabetes and concomitant hypertension have a high risk of developing macro- and microvascular complications. Early and effective treatment of hypertension in Type 2 diabetes is associated with a consistent reduction in macrovascular disease and a variable reduction in microvascular disease.

Diagnosis of hypertension

The revised value for the **diagnosis** of hypertension in patients with type 2 diabetes is:

BP > 140/80 mmHg on two separate days

The **revised BP target** for most persons with type 2 diabetes is between 120/70 and 140/80 mmHg (120/70 and ≤ 140/80)

Systemic hypertension is common in type 2 diabetes and will affect the majority of patients at some point in the course of their disease. Hypertension is an important modifiable risk factor for micro- and macrovascular disease.

Blood pressure (BP) must be measured at every clinic visit, after the patient has been seated and has rested for 5 minutes. It is essential to use an appropriately sized cuff, as small cuffs will yield falsely elevated BP readings.

Measure the BP in both arms at the initial consultation, and thereafter in the arm with the higher BP. The systolic BP should first be estimated by palpation and then auscultation, to avoid missing the auscultatory gap.

Ambulatory BP measurement (ABPM) should be utilised in selected cases when “white-coat hypertension” is suspected

Treatment of hypertension

Diabetes type 2 with/without evidence of microalbuminuria or proteinuria, start ACEi* as the treatment of choice, provided there are no contraindications.

If BP is not controlled on ACEi after one month despite adherence, titrate dose on a monthly basis to the maximum dose, until target BP is achieved.

Measure eGFR at initiation of treatment; 1–2 weeks after starting ACEi; and after dose titration.

If target BP is not achieved after 1 month on maximum dose of ACEi, add a diuretic.

If eGFR < 30 ml/min/1.73 m², do not use a thiazide diuretic.

If target BP is not reached after 1 month, despite adequate adherence, add a DHP CCB**.

If target BP is still not achieved, refer patient to hospital level.

If contraindication or intolerance to ACEi, use alternative agents.

* ACEi – angiotensin-converting enzyme inhibitor

** DHP CCB – dihydropyridine calcium-channel blocker (amlodipine, nifedipine)

Potential problems with certain antihypertensives

- Angiotension converting enzyme inhibitors (ACEi) may induce hyperkalaemia, renal failure, a persistent cough and lower glucose levels. Beware of angioedema and advise patients of the symptoms thereof (usually swelling around eyes and lips, life threatening airway obstruction can occur).
- Diuretics in large doses inhibit insulin release.
- Beta-blockers may mask symptoms of hypoglycaemia and exacerbate peripheral vascular disease
- Beta-blockers and diuretics may worsen dyslipidaemia

Additional treatment instructions for the healthcare provider

Renal function should be monitored. If potassium level is elevated, or creatinine level doubles, therapy with ACE inhibitors should be stopped and the patient referred for specialist investigation and management.

Check adherence - Refer Annexure F.

Treatment should be adjusted at each visit if BP is not at target (between 120/70 mmHg and 140/80 mmHg).

Any patient planning a pregnancy must be referred to a specialist for treatment changes – ACEi and diuretics should be stopped.

With a confirmed pregnancy, therapy should be switched to methyldopa and the patient should be referred to an antenatal unit equipped to manage high-risk pregnancies.

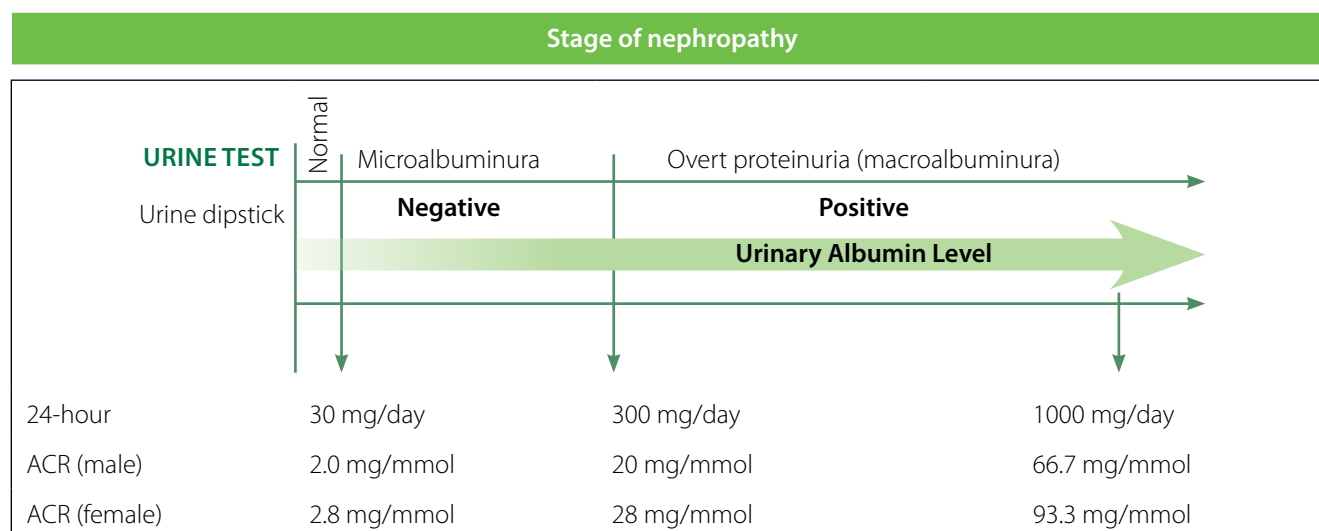
C5. CHRONIC KIDNEY DISEASE (CKD) /NEPHROPATHY - SCREENING AND MANAGEMENT ^{28,29,30,31,32}

Diabetic nephropathy is a frequent but potentially preventable long-term complication of diabetes mellitus, which is responsible for significant morbidity and mortality. Approximately 40% of patients with diabetes will develop CKD.

All patients diagnosed with Type 2 diabetes should be screened for CKD.

Before each screening and urine dipstick testing, transient causes of albuminuria (such as recent strenuous exercise, menstruation, fever, urinary tract infection, pregnancy, uncontrolled heart failure or acute severe elevation in blood pressure or blood glucose), low eGFR (such as dehydration, hypovolaemia) and acute renal failure on clinical grounds, should be excluded.

Classic diabetic nephropathy progresses from subclinical disease through to the earliest clinically detectable stage, which is characterized by persistent proteinuria.

Figure 1: Progression of classic diabetic nephropathy**Screening for CKD in diabetes: two tests are required**

- A random $\pm$ 20ml urine sample for determination of albumin-to-creatinine ratio (ACR).
- A 10ml non-fasting specimen of clotted blood for serum creatinine measurement, for conversion into the eGFR - patient's age, sex and race must be supplied for this purpose.

Frequency of screening

- If ACR value is within the normal range **and** eGFR > 60 ml/minute:
Screen annually.
- If the ACR is abnormal or the eGFR $\leq$ 60 ml/minute:
Repeat ACR twice and serum creatinine once within the next 3 months.

Management of CKD:

- Intensive diabetes management to achieve best possible glycaemic control for the prevention of onset and delay in progression to CKD.
- Control blood pressure and aggressively manage other *cardiovascular risk factors* (*smoking cessation and dyslipidaemia management*)
- Perform serum creatinine converted into an eGFR at least every 12 months to monitor disease progression and response to treatment.
- Adults with diabetes and persistent albuminuria (ACR > 2.0 mg/mmol in men; ACR > 2.8 mg/mmol in women) should be prescribed an ACE inhibitor (enalapril 10-20 mg daily; perindopril 2 mg daily; ramipril 2.5-10 mg daily and lisinopril 5-20 mg daily) to delay progression of CKD, even in the absence of hypertension.

NOTE: Coughing, the most common side-effect of ACE-inhibitors, may respond to a decreased dose or may spontaneously disappear after a few months. If intractable cough, refer patient.

- Patients should be monitored with follow-up serum creatinine and potassium levels within 1-2 weeks after initiation of ACE inhibitor or titration of dose, for significant worsening of renal function or for the development of hyperkalaemia. Serum creatinine may increase by up to 30% above baseline after initiation of an ACE inhibitor. If creatinine increases by > 20% from baseline, stop ACE inhibitor and consult a specialist.

When renovascular hypertension is suspected, patients should be referred to a physician experienced in renal medicine.

- A potassium restricted diet should be considered in the case of a mild to moderate increase in serum potassium concentration. Non-potassium sparing diuretics such as hydrochlorothiazide or furosemide may also be considered.

- Where eGFR is < 30 ml/minute, refer patient.
- ACE-inhibitors are contraindicated in pregnancy because of potential teratogenicity.

Referral for specialised renal care

Should be considered when:

- there is chronic progressive loss of renal function in spite of the suggested measures
- the eGFR is < 30 ml/min
- the ACR is persistently > 60 mg/ml
- BP targets cannot be reached
- patients cannot tolerate ACE inhibitors due to hyperkalaemia or a >20% increase in baseline serum creatinine within 3 months of initiating therapy.

Other causes of CKD in diabetes

More than 50% of people with diabetes and significant renal dysfunction, but with normal urinary albumin levels, have renal disease unrelated to classic diabetic nephropathy. A cause other than classic diabetic nephropathy should be considered under the following circumstances:

- The presence of extreme proteinuria (> 6 g/day)
- Persistent haematuria (micro- or macroscopic) or active urinary sediment
- Rapidly falling eGFR
- Low eGFR with little or no proteinuria
- Other complications of diabetes not present or not severe e.g. *diabetic retinopathy*
- Known duration of diabetes $\leq$ 5 years
- Family history of non-diabetic renal disease (e.g. polycystic kidney disease)
- Signs or symptoms of systemic disease

C6. DIABETIC RETINOPATHY (DR) ^{33,34,35,36}

Diabetic eye disease is a leading cause of visual impairment in the working-age population, and is mainly comprised of diabetic retinopathy and cataract formation. Diabetic retinopathy is preventable and/or treatable and cataract surgery is available, even though the numbers of operations performed have not achieved the South African CSR targets since 2005.

Stopping smoking and optimal control of blood glucose and blood pressure will substantially reduce the incidence of diabetic retinopathy.

Regular and thorough eye examination through dilated pupils and testing of visual acuity, should identify patients at risk of sight-threatening complications. These patients must be referred to an ophthalmologist.

Prevalence of diabetic retinopathy

Up to 40% of patients with diabetes have diabetic retinopathy while 8% have sight-threatening retinopathy. It is the responsibility of all health professionals to be vigilant and heed a patient's concern about his/her apparent eye problems.

The incidence of diabetes mellitus is increasing in the HIV population on ARVs and these patients therefore need to be screened regularly for possible eye complications.